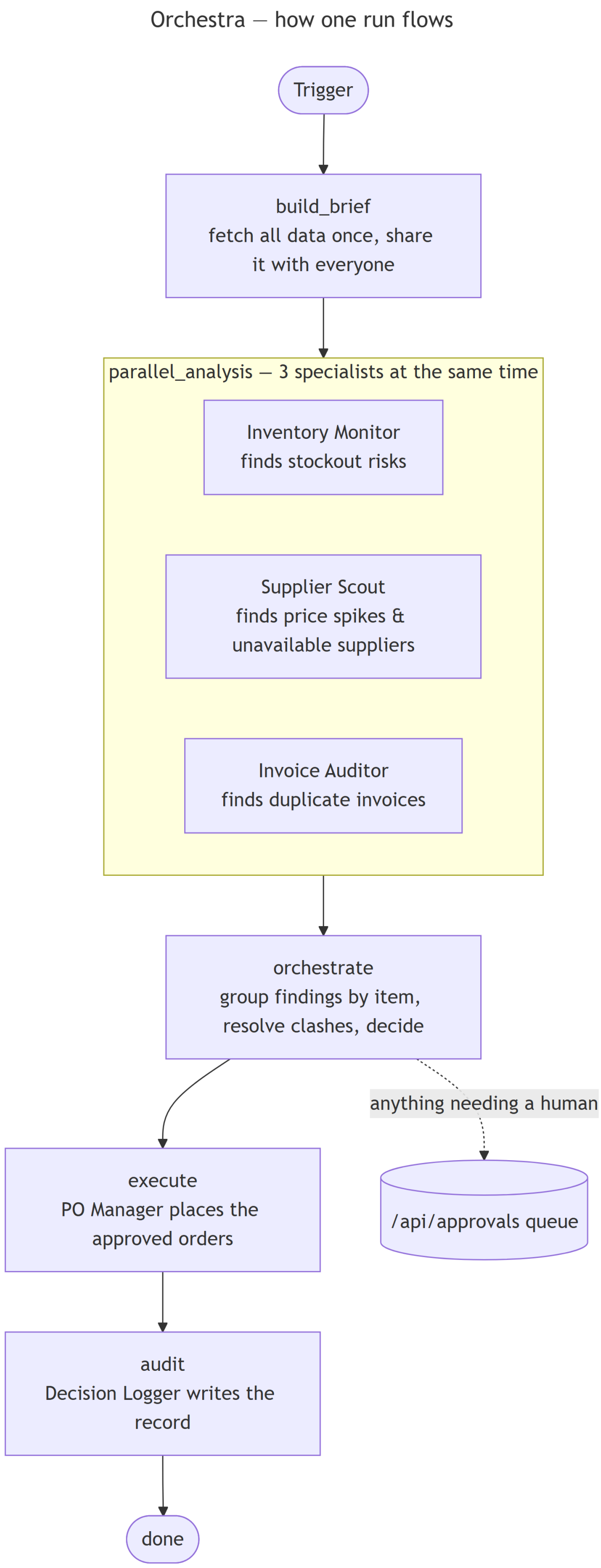
BETSY — THE ORCHESTRA

The second way Betsy can run: several specialists at once

WHAT THIS IS

The orchestra is Betsy's more advanced way of working. Instead of one checklist run top to bottom (that is the pipeline), three specialists look at the data at the same time, and a coordinator then pulls their findings together, settles any disagreements, and decides what to do.

It is useful when more than one thing is wrong with the same item at once — for example a part is running low AND its latest invoice looks like a duplicate. The pipeline would handle those one after another; the orchestra looks at them together and applies a clear rule for which one wins.



See the picture: diagrams/orchestra-overview.mmd

THE FIVE STEPS, IN PLAIN WORDS

1. BUILD BRIEF — gather the data once. Betsy pulls in stock, suppliers, purchase orders, and invoices a single time, then hands the same snapshot to all three specialists.

Because they all read the same fixed copy, they can safely run at the same time without tripping over each other.

2. PARALLEL ANALYSIS — three specialists work at once. The three specialists run together in parallel (see below). Each one reports back what it found. If one of them fails or takes too long, the others still finish and the run carries on with what is available.

3. ORCHESTRATE — pull it together and decide. The coordinator groups all the findings by item. If two findings clash on the same item, it applies a fixed rule to decide which one wins (see "How clashes are resolved"). It then turns each winning finding into a firm decision.

4. EXECUTE — carry out the approved decisions. A single component, the PO Manager, places the orders that were approved. It runs on its own, one at a time, so two orders can never be written at once. Anything that needs a person is left in the approvals queue instead.

5. AUDIT — write it down. The Decision Logger writes a record of the whole run. Like the pipeline, this step always runs, so every run leaves a trail.

THE THREE SPECIALISTS

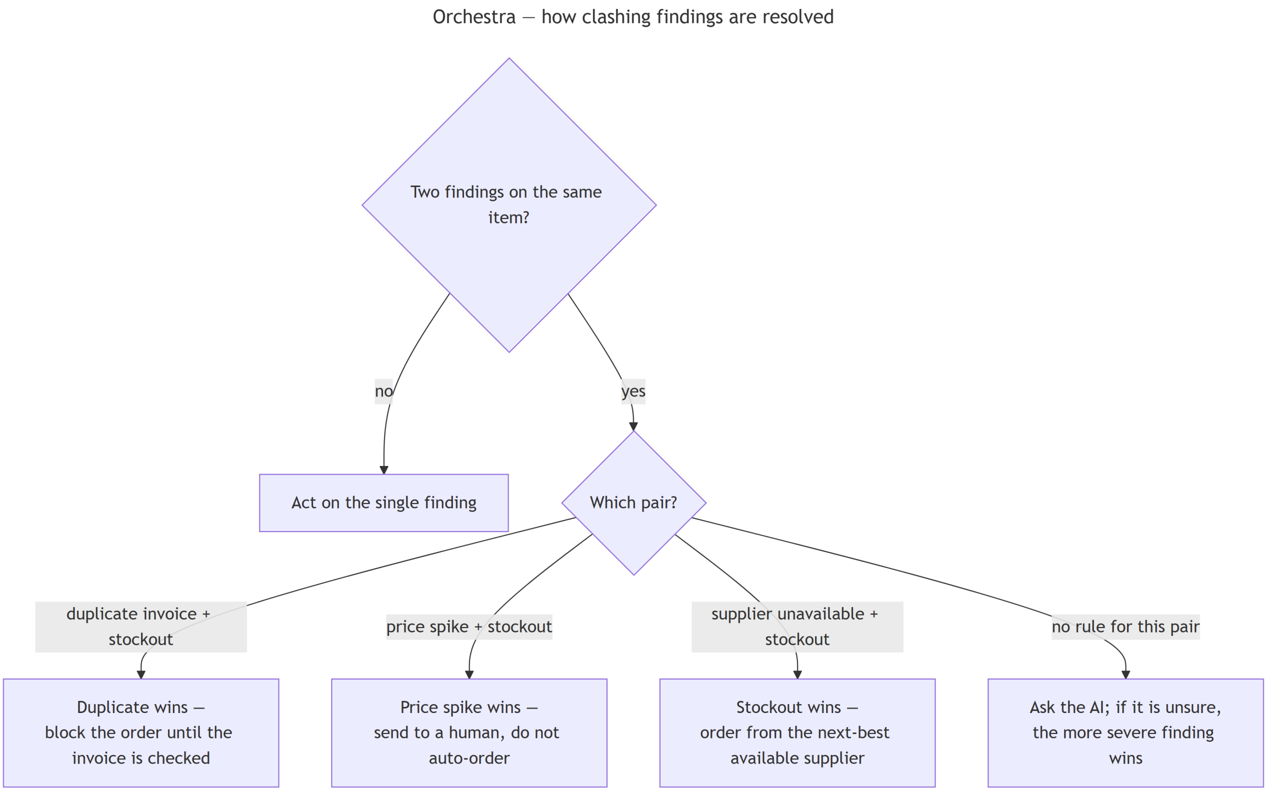
- Inventory Monitor looks at stock levels and flags items about to run out.
- Supplier Scout looks at supplier prices and availability, and flags price spikes or items no supplier can currently fill.
- Invoice Auditor looks at the invoices and flags likely duplicates.

Only one component ever writes anything back (the PO Manager, in the execute step). The three specialists only read, which is what makes it safe for them to run at the same time.

HOW CLASHES ARE RESOLVED

When two findings land on the same item, a short list of fixed rules decides which one wins, so the outcome is predictable:

- Duplicate invoice + stockout -> the duplicate wins. Betsy will not place an order on an item whose invoice is still in question.
- Price spike + stockout -> the price spike wins. The order is sent to a person rather than placed automatically.
- Supplier unavailable + stockout -> the stockout wins. Betsy orders from the next-best available supplier instead.
- Anything not covered by a rule -> Betsy asks the AI, and if the AI is unsure, the more severe finding wins.



See the picture: diagrams/orchestra-conflict.mmd

THE HUMAN CHECKPOINT

This works exactly as it does in the pipeline. The orchestra does not stop and wait for a person mid-run. Decisions that need a human are placed in the approvals queue, and Jenny approves or rejects them later from her screen. The approve/reject actions live in the server.

WHAT HAPPENS IF A STEP BREAKS

Each step records any problem instead of crashing the run, and the audit step always runs at the end. If one of the three specialists fails, the other two still report and the run continues with what they found.

THE MONEY SAFETY LIMIT

The same single limit applies: any order above the set ceiling (5,000 by default) is sent to a person instead of being placed automatically.

PIPELINE OR ORCHESTRA — WHEN EACH IS USED

The pipeline is the simpler choice and is fine for everyday runs where problems are handled one at a time. The orchestra is the choice when several problems can hit the same item at once and the order they are handled in matters. Both read and write through the same server and respect the same money limit and the same approvals queue.

WHY IT'S BUILT THIS WAY

The GAP analysis shows that real procurement problems do not arrive politely one at a time — a shortage, a price jump, and a questionable invoice can all involve the same part in the same week. The orchestra exists to handle that honestly: look at everything at once, then apply clear, written rules about which concern takes priority, so the result is both fast and defensible.

See: docs/gap-analysis.txt (and the styled version, bpm_analysis.html)

WHERE THIS LIVES IN THE CODE

- orchestra/graph.py the five steps, the clash rules, the decisions
- orchestra/state.py the shared snapshot passed between steps
- orchestra/run.py how an orchestra run is started
- orchestra/agents/inventory_monitor.py the stockout specialist
- orchestra/agents/supplier_scout.py the price/availability specialist
- orchestra/agents/invoice_auditor.py the duplicate-invoice specialist
- orchestra/agents/po_manager.py the only part that places orders
- orchestra/agents/decision_logger.py writes the run record
- server/routers/approvals.py the approve/reject screen actions
- shared/llm.py the AI connection (local Ollama, llama3.1:8b)

DIAGRAMS FOR THIS DOCUMENT

- diagrams/orchestra-overview.mmd the five steps, with the three specialists
- diagrams/orchestra-conflict.mmd how clashing findings are resolved